ENTERPRISE TELEPHONY & REVENUE PLATFORM

Complete CRM System

Product Overview, Architecture & Operational Workflow

A comprehensive operational and architectural handbook for Development Teams, Management, HR, Sales, and Support. Covering automated lead capture, intelligent assignment, Doocti CTI telephony integration, role-based governance, and real-time revenue analytics.

TARGET AUDIENCE
Dev, Leadership, Sales, HR & Support

TECH STACK
React 18 + FastAPI + SQLite/MySQL + Doocti

VERSION & SCOPE
2026 Production & Mobile SIM Roadmap

EXECUTIVE CONTEXT

1. Introduction to the CRM Platform

Why our organization requires an integrated, high-velocity CRM system

What is CRM?

● Core Foundation

• **Definition:** Customer Relationship Management (CRM) is a centralized software ecosystem managing every customer and prospect interaction across their entire lifecycle.

• **Beyond Contact Storage:** It is not just a digital address book; it is an active revenue engine connecting marketing channels, telephony trunks, sales reps, and post-sale support.

• **Single Source of Truth:** Consolidates omnichannel customer touchpoints—website forms, Meta Facebook ads, phone calls, WhatsApp messages, and quotations—into unified customer profiles.

• **Real-time Synchronization:** Eliminates departmental silos by instantly reflecting deal progress, call logs, and customer follow-up notes across all teams.

Why Our Company Needs This CRM

● Business Imperative

• **Eliminating Lead Leakage:** Incoming inquiries from web forms and paid Meta ads previously suffered manual transfer delays, causing high drop-offs. CRM automates immediate ingestion.

• **Speed-to-Lead Acceleration:** Studies show calling a lead within 5 minutes increases conversion by 391%. Integrated Doocti telephony allows instant 1-click dialing.

• **Transparent Accountability:** HR and Leadership gain objective visibility into employee call volumes, talk time, missed call callbacks, and conversion ratios.

• **Data Security & Ownership:** Prevents customer lead theft by restricting raw contact exports, masking contact numbers where needed, and requiring admin approval for deletions.

Core Objectives & ROI

● Expected ROI

• **Sales Velocity:** Empowers sales executives to handle 60-80 calls daily with automated dialers and rapid disposition logging.

• **Lead Re-Inquiry Intelligence:** Automatically detects returning prospects, links new inquiries to existing profiles, and alerts the assigned executive.

• **Structured Hierarchy:** Multi-tier role model (Super Admin, Admin, Manager, Executive) ensures team leaders mentor reps without data conflicts.

• **Measurable Revenue Growth:** Directly links call effort, meeting scheduling, and quotation submissions (QuoteTask integration) to closed-won revenue.

REAL-TIME VISIBILITY

2. CRM Operational Dashboard & Analytics

Instant KPIs, pipeline distribution, and telephony volume metrics

TOTAL LEADS
Pipeline Aggregate
All captured leads across all time and channels with live filtering

NEW LEADS
Actionable Inflow
Fresh leads received today requiring immediate first-contact dialing

FOLLOW-UP LEADS
Active Pipeline
Leads with scheduled callbacks, client consultations, or demo sessions

CONVERTED LEADS
Revenue Won
Successfully closed deals transitioned to active customer accounts

Pipeline Status Breakdown & Flow

- **Lead Status Lifecycle**
- **Assigned Leads:** Leads successfully routed to sales executives with pending initial contact.
- **Pending Leads:** Unclaimed or awaiting allocation based on department capacity rules.
- **Follow-up Leads:** Scheduled callbacks with date/time pickers and alarm notifications.
- **Converted (Won):** Completed sales transitions verified with proforma quotation links.
- **Rejected / Lost:** Categorized with structured loss reasons (Budget, Competitor, Invalid Number).
- **Click-to-Filter:** Clicking any KPI card instantly filters the lead table below without page reloads.

Telephony Metrics & Rep Leaderboard

- **Voice & Effort Telemetry**
- **Today's Total Calls:** Aggregated inbound and outbound telephony interactions processed by Doocti.
- **Missed Calls Tracker:** Critical KPI tracking unreturned customer calls with 1-click redial alerts.
- **Inbound vs. Outbound Split:** Visual ratio tracking proactive sales reach-outs vs. customer-initiated calls.
- **Call Duration Analytics:** Average talk time per rep highlighting high-engagement consultations.
- **Employee Leaderboard:** Live ranking of top performers based on call volume, talk time, and deals closed.
- **Date & Hierarchy Filters:** Managers can filter by Today, Yesterday, This Week, or specific team members.

CORE MODULE

3. Complete Lead Management Lifecycle

Omnichannel lead ingestion, rich profiles, re-inquiry tracking, and data actions

Omnichannel Lead Ingestion

- **Capture Channels**

• **Manual Lead Creation:** Fast single-lead entry modal for walk-ins, phone inquiries, and direct referrals with validation.

• **Bulk Excel / CSV Import:** Standardized spreadsheet importer with column validation, error logging, and duplicate phone skipping.

• **Website Webhook Capture:** Direct JSON API capturing landing page submissions with UTM campaigns, landing URL, and product interest.

• **Meta / Facebook Ads API:** Automated webhook ingestion syncing Facebook Lead Gen forms directly into CRM within seconds.

• **Source Tagging:** Every lead retains pristine attribution (Website, Meta Ads, Excel, Manual, WhatsApp).

Customer Profile & Status Flow

- **Data Architecture**

• **Comprehensive Records:** Stores full name, phone, WhatsApp number, email, company, industry, city, state, and notes.

• **Product Categorization:** Categorizes requirements (e.g. CRM, ERP, Cloud Telephony) with confirmation flags.

• **Status Lifecycle:** Tracks state transitions: NEW -> ASSIGNED -> CONTACTED -> FOLLOW_UP -> QUOTED -> WON / LOST.

• **Re-Inquiry Engine:** Intelligently flags returning customers (`is_reinquiry=True`), increments inquiry count, and logs previous sources.

• **QuoteTask Integration:** Maps quotations directly to leads with status, quote amount, and PO tracking.

Data Governance & Filtering

- **Control & History**

• **Fast Search & Sorting:** Instant search across customer name, phone number, company, email, or city.

• **Multi-Parameter Filtering:** Filter by Date Range, Product Category, Assigned Rep, Status, Priority, and Source.

• **Audit History & Activity Log:** Every field edit, reassignment, status transition, and call is permanently logged in `lead_activities`.

• **Controlled Deletions:** Prevents accidental data loss; delete requests require Super Admin / Admin review and approval.

• **Export Governance:** Only authorized managerial roles can export lead datasets to Excel.

OPERATIONAL EFFICIENCY

4. Intelligent Automatic Lead Assignment

Automated distribution rules, managerial hierarchies, and workload balancing

Automated Distribution Rules

- **Routing Logic**

• **Round-Robin Engine:** Evenly distributes incoming leads in cyclic order across active team members within a department.

• **Workload-Based Balancing:** Calculates pending lead quotas per executive; prioritizes reps with lower active workloads.

• **Role & Department Specific:** Directs enterprise software inquiries to Senior Sales reps, and service tickets to Support.

• **Active Status Awareness:** Only allocates leads to employees currently marked 'Active' and clocked in for duty.

Manager Hierarchy & Manual Overrides

- **Hierarchy Control**

• **Managerial Routing:** Leads can be routed to Sales Managers who delegate to their specific sub-teams.

• **Manual Reassignment:** Team leaders and Admins can reassign leads at any time with structured reassignment notes.

• **Reassignment Audit Trail:** System logs `assigned_by`, `assigned_to`, timestamp, and method ('AUTOMATIC' vs 'MANUAL').

• **Notification Dispatch:** Executives receive instant toast notifications and dashboard alerts upon receiving new leads.

Duplicate Prevention & Integrity Safeguards

- **Data Integrity Engine**

• **Phone Number Uniqueness Check:** Before creating a lead, system cleanses phone numbers (stripping country codes, spaces, special chars) and checks database indexes to prevent identical records.

• **Re-Inquiry Attachment:** If an existing phone number submits a new inquiry form, the assignment engine can either keep the original account manager or notify the manager for reallocation based on company policy.

• **Ghost Ingestion Blocker:** Rejects empty or test submissions before assignment, preserving employee quotas and productivity.

• **Race Condition Prevention:** Database row locks and unique constraint indexing ensure concurrent webhook hits cannot generate duplicate lead assignments.

HUMAN CAPITAL

5. Employee Management & Organizational Structure

User lifecycle, telephony station mapping, reporting lines, and productivity

Employee Profiles & Lifecycle

• HR Operations

• **Profile Registration:** HR or Admin creates employee records with corporate email, full name, phone number, and employee code.

• **Department Assignment:** Assigned to dedicated units: Sales, Business Development, Customer Support, IT, or Management.

• **Designation Structure:** Configured titles (e.g. Sales Executive, Team Lead, Sales Manager, VP of Revenue).

• **Lifecycle Status:** One-click toggle between Active and Inactive status; deactivating an employee immediately revokes system access and suspends lead routing.

Telephony Station & ID Mapping

• CTI Integration

• **Extension Provisioning:** Each sales and support executive is mapped to a dedicated PBX Station ID (e.g. Extension 1282).

• **SIP Agent Credentialing:** Links CRM user account to Doocti PBX username (`chntechnolog_agent01`) for softphone authentication.

• **Inbound Routing Linkage:** When an incoming call arrives at station 1282, CRM automatically identifies the answering employee.

• **Device Flexibility:** Configurable for WebRTC browser dialer or mobile client without re-registering employee identities.

Reporting Hierarchy & Metrics

• Performance Management

• **Reporting Manager Tree:** Employees are mapped to their specific reporting manager (`manager_id` foreign key).

• **Scoped Visibility:** Managers automatically view leads, calls, and performance metrics of their direct subordinates.

• **Activity Monitoring:** Monitors login timestamps, last active interaction, daily completed calls, and follow-up compliance.

• **Employee-Wise Reports:** Generates individual performance scorecards comparing leads assigned, conversion rates, and revenue generated.

GOVERNANCE & SECURITY

6. Role-Based Access Control (RBAC)

Hierarchical authorization matrix enforcing least-privilege enterprise security

Super Admin

Full Platform Sovereign

Unrestricted system access: database configuration, billing, system audit logs, API credentials, and role permission overrides.

Admin

Operational Administrator

Manages employee accounts, integration settings, master lead assignments, bulk data exports, and deletion approval requests.

Sales Manager

Team Pipeline Commander

Views entire team's leads, monitors subordinate call logs, reassigns leads, reviews deal pipelines, and accesses team sales reports.

Sales Executive

Individual Contributor

Views and manages strictly assigned leads; initiates calls, updates notes, schedules follow-ups, and requests quotes.

Granular Permission Capabilities Across Core Modules

- **RBAC Security Matrix**
- **Lead Operations:** View All Leads (Admin/Manager), View Assigned Only (Executive), Create Lead, Edit Lead Details, Reassign Leads.
- **Data Protection & Deletion:** Hard delete is restricted. Deletion requests are logged in `deletion_requests` table requiring Admin review.
- **Call Logs & Telephony:** Executives view only their own call history; Managers inspect team recordings; Admins audit total PBX traffic.
- **Export Governance:** Only Admins and authorized Managers can trigger Excel/CSV downloads, protecting company customer intelligence.
- **Configuration Access:** Doocti settings, Meta Webhook tokens, and WhatsApp API keys are isolated to Admin and Super Admin roles.
- **Audit Trail Compliance:** All role modifications and unauthorized access attempts are logged into `audit_logs` for compliance review.

LIVE TELEPHONY INFRASTRUCTURE

7. IVR & Doocti Telephony Integration

Production WebRTC dialer, DID routing, caller ID lookup, and webhook ingestion

WebRTC Softphone & Click-to-Call

• Softphone Architecture

• **Embedded Softphone:** In-app WebRTC dialer iframe (`beta3-app.doocti.com`) seamlessly integrated into the CRM navigation bar.

• **1-Click Click-to-Dial:** Clicking any lead phone number dispatches an HTML5 `postMessage` event to the Doocti softphone to dial instantly.

• **No External Deskphones:** Eliminates expensive hardware; reps make high-fidelity voice calls directly through browser headsets.

• **DID Inbound Routing:** Company DID numbers map to interactive IVR queues, routing incoming callers to available sales agents.

Inbound Screen Pop & Caller ID

• Inbound Intelligence

• **Instant Caller Identification:** When an inbound call rings, Doocti queries CRM API by incoming phone number to fetch caller records.

• **Screen Pop-Up Modal:** CRM displays customer name, existing requirements, previous interaction notes, and assigned executive.

• **Warm Customer Greeting:** Sales reps answer calls knowing the customer's full history, significantly improving customer experience.

• **Missed Call Alerts:** If an inbound call goes unanswered, CRM logs a `MISSED` call event and alerts the team for rapid callback.

Real-time CDR Webhook & Audio

• Webhook Data Stream

• **Production Webhook:** Doocti PBX pushes Call Detail Records (CDR) to `/api/doocti/call-log` immediately after call hangup.

• **Normalized Call Data:** Captures unique session ID, caller number, agent station, call status, duration, and timestamps.

• **Call Audio Recordings:** Doocti pushes secure cloud storage URLs (`recording_path`) for immediate in-browser playback.

• **Ghost Call Rejection:** CRM filters out station self-dial loops and internal PBX test events to keep analytics 100% accurate.

PLANNED MOBILE ARCHITECTURE

8. Mobile SIM Calling Integration – Android

Field sales calling via mobile SIM, call detection, duration tracking, and privacy

Planned Android Mobile App

- **Mobile App Roadmap**

- **Native Flutter Framework:** Cross-platform mobile client (`mobile/` directory) designed for on-the-field sales executives.
- **Direct SIM Dialing:** Allows reps to trigger calls using their physical device SIM card when away from office desktops.
- **Call Intent Integration:** Initiates Android native dialer intent pre-filled with customer phone number from CRM lead card.
- **Seamless Offline Queue:** Caches lead interactions locally when network coverage is weak and syncs upon reconnection.

Call Detection & Metadata Sync

- **Metadata Telemetry**

- **Telephony State Detection:** Listens to Android `TelephonyManager` call state transitions (CALL_STATE_RINGING, OFFHOOK, IDLE).
- **Call Log Permission Access:** Queries native Android CallLog provider with user permission to calculate precise call duration.
- **Automatic Backend Sync:** Sends call metadata (employee ID, lead phone, call type, start time, duration) to `/api/mobile/call-logs`.
- **Lead History Linkage:** Backend automatically attributes the mobile SIM call to the respective customer activity timeline.

Critical Android Technical & Privacy Boundaries (Important Distinction)

- **Technical Reality & Policy Compliance**

- **Android OS Audio Recording Constraints:** Modern Android versions (Android 10 through 14+) strictly block third-party background call audio recording at the Linux kernel and framework levels to protect privacy.
- **No Silent Background Audio Recording:** Our CRM mobile app DOES NOT claim or attempt unrestricted background call voice recording. It strictly captures call telemetry (timestamp, duration, recipient, call status).
- **Strict Permission Governance:** Requires explicit runtime user consent for `READ_PHONE_STATE`, `CALL_PHONE`, and `READ_CALL_LOG` permissions in compliance with Google Play Store Developer Policies.
- **Hybrid Enterprise Architecture:** Office reps use WebRTC Doocti for full audio recording and CTI; field reps use SIM calling with automated metadata duration logging.

TELEPHONY GOVERNANCE

9. Comprehensive Call Logs Management

Multi-status disposition, audio recording playback, and real-time updates

Call Status Classification

• **Call Dispositions**

• **Answered / Connected:** Successful conversation where billable duration was logged by the PBX.

• **Missed / No Answer:** Unanswered inbound customer calls or unanswered outbound sales attempts.

• **Busy / Engaged:** Carrier signal indicating line occupied; automatically schedules quick callback.

• **Failed / Rejected:** Carrier routing errors or user-rejected calls logged for diagnostic review.

• **Status Normalization:** Raw PBX dispositions (`COMPLETED`, `NOANSWER`, `CHANUNAVAIL`) are mapped to clean CRM statuses.

Log Metrics & Audio Playback

• **Audio & Telemetry**

• **Detailed Call Metrics:** Displays Call Unique ID, Customer Phone, Employee Name, Extension Station, Duration (seconds), and Date/Time.

• **Lead Association:** Automatically associates call records with matching CRM leads, updating the customer's last-contacted date.

• **Direct Audio Player:** Integrated HTML5 audio wave player allowing managers to listen to recorded calls with seek and speed controls.

• **Quality Audit & Coaching:** Enables sales coaches to evaluate pitch quality, customer objections, and script adherence.

Real-Time WebSocket Sync

• **Real-Time UI Engine**

• **No Page Refresh Required:** Integrated WebSocket service (`backend/websocket/telephony.py`) pushes new call events instantly to browser clients.

• **Live Inbound Banner:** When a call completes, the call logs table prepends the record with visual highlight animations.

• **Multi-Parameter Filtering:** Filter logs by Employee, Call Direction (Inbound/Outbound), Status, Duration Range, and Date Range.

• **Managerial Export:** Allows authorized supervisors to export filtered telephony logs to Excel for executive reporting.

MARKETING AUTOMATION

10. Meta / Facebook Lead Ads Integration

Direct Graph API webhook, instant lead ingestion, field mapping, and deduplication

Meta Graph API Architecture

• API Configuration

• **Meta Developer App:** Enterprise Facebook application configured with `leads_retrieval` and `pages_manage_ads` permissions.

• **Page Access Tokens:** Secure encrypted storage of long-lived Page Access Tokens linking corporate Facebook business pages.

• **Webhook Subscription:** CRM subscribes to Meta's `leadgen` real-time webhook topic for immediate event delivery.

• **Cryptographic Verification:** Validates Meta `X-Hub-Signature-256` HMAC signatures using the App Secret to block spoofed payloads.

Field Extraction & Mapping

• Payload Mapping

• **Automated Lead Ingestion:** When a user submits a Facebook Instant Form, Meta dispatches `leadgen_id` to our webhook.

• **Graph API Query:** CRM queries `graph.facebook.com/v19.0/{leadgen_id}` to retrieve user responses in real-time.

• **Intelligent Field Mapping:** Maps standard fields (Full Name, Phone, Email, City) and custom form questions (Budget, Timeline, Service).

• **Source & Campaign Attribution:** Captures Facebook Page Name, Ad ID, AdSet ID, and Campaign Name into the lead record.

Deduplication & Error Recovery

• Pipeline Integrity

• **Duplicate Lead Checking:** Checks for existing leads matching the phone number before creating a new database record.

• **Re-Inquiry Flagging:** If the lead already exists, records an inquiry touchpoint rather than generating duplicate cards.

• **Immediate Auto-Assignment:** Newly ingested Meta leads immediately pass to the auto-assignment engine for instant rep allocation.

• **Resilient Error Handling:** Payloads that fail schema parsing are stored in an integration dead-letter queue for admin reprocessing.

INBOUND CHANNEL

11. Website Lead Ingestion Engine

REST API webhooks, website contact form capture, validation, and auto-routing

Website Contact Form Webhook

- **Ingestion Endpoint**

• **Dedicated REST Endpoint:** Exposes `/api/website/leads` to accept incoming inquiry payloads from corporate landing pages.

• **Universal Form Support:** Compatible with WordPress (Contact Form 7, Elementor), Webflow, Custom HTML/React forms, and PHP sites.

• **Payload Flexibility:** Accepts standard JSON bodies and URL-encoded form submissions from server-side handlers.

• **API Key Security:** Protected by pre-shared header tokens (`X-Website-Token`) ensuring only authorized web assets submit data.

Data Validation & Sanitization

- **Sanitization Pipeline**

• **Phone Normalization:** Strips non-numeric characters, validates 10-digit mobile standards, and standardizes country prefix (+91).

• **Email RFC Validation:** Validates syntax and rejects disposable or malformed email addresses before saving.

• **Required Field Verification:** Ensures contact name, mobile number, and requirement summary are populated before ingestion.

• **Spam / Bot Filtering:** Honeypot field detection and IP rate limiting to block automated website spam submissions.

Attribution, Auto-Assignment & Instant Notifications

- **Real-Time Processing**

• **Full Campaign Attribution:** Captures source landing page URL, referring domain, UTM source, UTM medium, and UTM campaign parameters.

• **Instant Auto-Assignment:** Pushes the validated lead into the auto-assignment engine within milliseconds of form submission.

• **Immediate Sales Notification:** Assigned sales rep receives an instant desktop pop-up notification with customer phone and requirement summary.

• **Automated Welcome Trigger:** Can trigger an automated WhatsApp or email acknowledgment welcoming the customer and confirming their inquiry.

CUSTOMER MESSAGING

12. WhatsApp Business Integration

Pre-approved templates, transactional alerts, and planned 2-way conversation chat

WhatsApp Business Platform

● API Infrastructure

• **Enterprise Cloud API:** Connected via official WhatsApp Business Platform / Meta Cloud API and authorized BSP gateways (e.g. Gupshup).

• **Verified Business Profile:** Sends messages from our verified corporate WhatsApp number with brand identity and green badge.

• **Template Message Engine:** Manages pre-approved Meta message templates for outbound transactional and notification triggers.

• **Variable Dynamic Merging:** Automatically populates template parameters (Customer Name, Quotation Number, Meeting Time, Rep Name).

Implemented Notification Triggers

● Live Features (Implemented)

• **Lead Welcome Notification:** Instant automated greeting sent to new leads confirming receipt of their inquiry.

• **Quotation Delivery Alert:** Sends quotation PDF link and summary when a formal estimate is generated via QuoteTask.

• **Follow-Up Reminder:** Sends appointment confirmations and reminder alerts to prospects before scheduled meetings.

• **Delivery Status Tracking:** Tracks message statuses where supported (Sent, Delivered, Read, Failed) in the CRM communication log.

Planned Interactive Capabilities

● Roadmap (Planned)

• **Bidirectional 2-Way Chat UI:** Planned in-app chat inbox allowing sales reps to converse directly with leads via WhatsApp inside CRM.

• **Document & Media Exchange:** Enables customers to send brochures, architectural plans, and payment receipts directly to the CRM card.

• **Interactive Quick Replies:** Automated buttons allowing leads to click 'Confirm Meeting', 'Request Callback', or 'Talk to Human'.

• **Clear Feature Separation:** Outbound template alerts are currently operational; interactive 2-way chat console is on the active roadmap.

PIPELINE NURTURING

13. Follow-Up & Reminder Management

Scheduling consultations, agenda tracking, reminder notifications, and history

Follow-Up Scheduling Engine

• Scheduling Logic

• **Precise Date & Time Picker:** Sales executives schedule callbacks with exact appointment date and time selection.

• **Structured Agendas:** Categorizes follow-up purposes: First Demo, Price Negotiation, Technical Review, Executive Decision.

• **Contextual Notes:** Reps log specific discussion points from the previous call to ensure continuity in future interactions.

• **Lead Status Synchronization:** Setting a follow-up automatically updates the lead status to `FOLLOW_UP`, keeping pipelines organized.

Queues & Daily Action Lists

• Execution Queues

• **Pending Follow-ups Queue:** Dedicated view displaying all callbacks scheduled for today that require execution.

• **Overdue Follow-ups Tracker:** Highlights missed or overdue follow-ups with high-visibility red badges to prevent customer drop-off.

• **Completed Follow-ups:** Historical archive of executed appointments linked with call logs and updated lead stages.

• **Daily Target Alignment:** Enables sales reps to structure their day into morning new lead dialing and afternoon scheduled consultations.

Reminders & Managerial Oversight

• Governance & Continuity

• **Automated Reminders:** System notification service (`followup_reminder_service.py`) triggers browser toasts 15 minutes before appointments.

• **Lead Activity Timeline:** Every scheduled, rescheduled, and completed follow-up is permanently appended to `lead_activities`.

• **Managerial Compliance Audit:** Sales managers inspect subordinate follow-up completion rates during weekly pipeline reviews.

• **Customer Journey Continuity:** If a rep takes leave, their scheduled follow-ups can be reassigned seamlessly without losing context.

BUSINESS INTELLIGENCE

14. Reports, Analytics & Executive Dashboards

Multi-dimensional performance reports, conversion velocity, and data exports

Sales & Pipeline Analytics

● Revenue Metrics

• **Daily Lead Inflow Reports:** Visual breakdown of new leads captured per day segmented by acquisition source.

• **Conversion Funnel Analysis:** Tracks progression rates from New -> Contacted -> Follow-up -> Quoted -> Won.

• **Lead Loss Reasons Analysis:** Aggregates rejection categories (Pricing, Unresponsive, Competitor) to guide product and pricing adjustments.

• **Sales Targets vs. Achievement:** Monitors team and rep performance against monthly revenue and closed-deal targets (`sales_targets.py`).

Telephony & Rep Productivity

● Productivity Telemetry

• **Call Volume Scorecards:** Compares daily call attempts, answered calls, and talk time across all sales executives.

• **Inbound vs. Outbound Effort:** Measures proactive outreach vs reactive inquiry handling across business development teams.

• **Missed Call Recovery Velocity:** Tracks average minutes elapsed before an executive returns an unanswered inbound customer call.

• **Employee Activity Timeline:** Visual audit of daily user actions (logins, calls, notes logged, status changes) for HR performance reviews.

Channel ROI & Data Export

● Governance & Export

• **Lead Source ROI:** Compares cost-per-lead and conversion percentage across Meta Ads, Website SEO, Walk-ins, and Referrals.

• **Excel & CSV Data Export:** Secure export module (`DataExportPage.jsx`) enabling managers to download filtered lead and call datasets.

• **Export Permission Restriction:** Hardened RBAC restricts export access to prevent unauthorized bulk customer list exfiltration.

• **Audit Logging for Exports:** Every data export event logs the requesting user ID, timestamp, and query filter parameters.

SYSTEM ADMINISTRATION

15. Admin Settings & Integrations Configuration

Centralized administrative cockpit for telephony, API keys, and system parameters

Doocti Telephony Settings

● Telephony Gateway

• **WebPhone URL:** Configurable softphone iframe host URL (`https://beta3-app.doocti.com/`).

• **WebRTC Domain & WSS:** PBX SIP signaling domain and WebSocket secure endpoint configuration.

• **Tenant & Organization ID:** Multi-tenant company identifier (`chntechnolog`) segregating telephony trunks.

• **DID Number Management:** Configures inbound DID trunks and sets fallback destination queues.

• **Extension Mapping Matrix:** Direct administrative mapping between employee CRM profiles and PBX extension station IDs.

Meta & Website Settings

● API Management

• **Meta App Configuration:** Manages Facebook App ID, App Secret, and Webhook Verification Tokens.

• **Page Token Management:** Interface to paste and refresh long-lived Facebook Page Access Tokens.

• **Lead Form Mapping:** Configures custom field mapping between Facebook Instant Form field names and CRM attributes.

• **Website Webhook Secrets:** Generates pre-shared authentication keys for landing page webhook endpoints.

• **Test Payload Dispatcher:** Built-in integration sandbox to simulate incoming test leads and verify parsing.

Governance & Deletion Queue

● Security & Governance

• **Company Profile & Hours:** Sets organization details, working days, business hours, and timezone (IST standard).

• **Deletion Request Queue:** Admin approval console (`DeleteRequestsPage.jsx`) where soft-deleted leads await final review.

• **Role Permission Editor:** Interactive permission matrix to toggle granular module rights per role.

• **Notification Preferences:** Configures automated email and WhatsApp alert triggers across system workflows.

TECHNICAL ARCHITECTURE

16. End-to-End Database & System Architecture

Multi-layer enterprise stack connecting clients, API services, database, and CTI trunks

1. CLIENT PRESENTATION LAYER

React 18 Web Application (Vite, Axios, WebSockets) | Embedded Doocti WebRTC Softphone Iframe | Flutter Mobile App (Planned)

2. API GATEWAY & APPLICATION LOGIC

FastAPI (Python 3.12/3.14) | Uvicorn ASGI | JWT Authentication | RBAC Middleware | Real-time Telephony WebSocket Hub

3. DATA STORAGE & PERSISTENCE LAYER

SQLAlchemy 2.0 ORM | MySQL (Production) / SQLite (Dev) | Tables: leads, employees, call_logs, followups, activities, audit_logs

4. EXTERNAL TELEPHONY & INTEGRATION SERVICES

Doocti Cloud PBX (Asterisk/SIP) | Meta Graph API Webhooks | Website Lead API | WhatsApp Cloud API | QuoteTask Service

PROCESS ENGINEERING

17. End-to-End Lead-to-Sales Workflow

The 10-stage customer journey from initial inquiry generation to closed-won revenue

1. Generation

Meta Ads, Website forms, Excel upload, or direct referral.

2. Ingestion

FastAPI webhook receives JSON, validates data, and blocks duplicates.

3. Assignment

Auto-assignment engine routes lead to sales rep based on workload.

4. Notification

Sales executive receives instant toast alert and dashboard entry.

5. 1st Contact

Rep clicks phone icon; Doocti softphone dials customer within minutes.

6. Engagement

Rep logs call disposition (Answered, Busy) and discussion notes.

7. Nurturing

Rep schedules follow-up appointments with reminder alarms.

8. Quotation

Generates formal proposal via QuoteTask; sends WhatsApp alert.

9. Deal Won

Customer accepts; deal converted to active Customer account.

10. Analytics

Management reviews conversion velocity and rep performance scorecards.

STANDARD OPERATING PROCEDURE

18. Sales Executive Daily Operational Workflow

Step-by-step daily cadence for high-performance sales execution

09:00 AM - Login & Extension Check | Log in to CRM, open the embedded Doocti softphone, verify green 'Registered' station status, and review assigned target quota.

09:15 AM - Review Today's Follow-ups | Navigate to 'Followups' tab; inspect pending callbacks, scheduled client demos, and overdue reminders requiring immediate action.

10:00 AM - Work New Incoming Leads | Review freshly assigned leads from overnight Meta Ads and Website campaigns. Execute speed-to-lead 1-click dialing.

11:30 AM - Log Dispositions & Notes | Immediately after each call, log accurate disposition, update customer requirement notes, and categorize product interest.

02:00 PM - Afternoon Follow-Up Dialing | Execute scheduled consultations, deliver tailored pitches, and send pre-approved WhatsApp message templates.

04:30 PM - Quotations & Deal Closures | Coordinate with QuoteTask integration to issue proforma quotations; initiate Deal Won conversion workflows.

06:00 PM - Daily Wrap-up & Target Audit | Review personal metrics on the Dashboard: total calls placed, talk time, follow-ups completed, and next-day scheduled tasks.

MANAGEMENT CADENCE

19. Sales Manager Workflow & Pipeline Governance

Supervising team activity, rebalancing workloads, and coaching for conversion

Team Activity & Telephony Audit

● Daily Supervision

• **Real-time Team Monitoring:** Supervises live call activity and extension status of all subordinate sales representatives.

• **Daily Talk Time Audits:** Inspects total billable minutes and outbound call volume to ensure rep outreach benchmarks are met.

• **Call Audio Coaching:** Samples recorded calls from Doocti storage to provide constructive coaching on pitch and objection handling.

• **Missed Call Tracking:** Monitors team response time to missed inbound customer inquiries.

Pipeline & Allocation Control

● Pipeline Governance

• **Lead Rebalancing:** Reallocates stale or untouched leads from underperforming or overloaded reps to active top performers.

• **Quota & Target Tracking:** Sets and tracks monthly sales quotas, monitoring individual percentage-to-goal achievement.

• **Follow-Up Discipline:** Reviews the Overdue Follow-ups queue, holding reps accountable for missed client appointments.

• **Conversion Funnel Analysis:** Identifies bottlenecks where leads stall (e.g. Quoted but not closing).

Reporting & Business Review

● Executive Reporting

• **Executive Team Reports:** Generates weekly sales performance digests for executive leadership review.

• **Source Performance Feedback:** Provides direct feedback to marketing regarding lead quality by campaign source.

• **High-Value Deal Approvals:** Reviews and signs off on custom discounts and enterprise quotation terms.

• **Support Escalations:** Seamlessly routes post-sale requirements to Support Managers via integrated workflow queues.

PLATFORM ADMINISTRATION

20. Super Admin & Operational Administrator Workflow

Global platform governance, integration health, data sovereignty, and security

User & Role Governance

● Identity Management

• **Employee Onboarding:** Provisions new user accounts, assigns departmental roles, and configures reporting managers.

• **Telephony Extension Assignment:** Allocates Doocti PBX station IDs and SIP credentials to newly registered reps.

• **Role Matrix Customization:** Toggles granular permission flags across departments in `RoleManagementMatrix.jsx`.

• **One-Click Offboarding:** Instantly revokes access for departed personnel and safely reassigns their active pipelines.

Integration Health & API Audits

● Integration Health

• **Doocti PBX Connectivity:** Monitors webhook ingestion health at `/api/doocti/call-log` to ensure zero dropped call logs.

• **Meta Webhook Verification:** Inspects Facebook App subscriptions, monitors token expiration dates, and reviews error logs.

• **Website Webhook Secrets:** Issues and rotates security keys for public website landing page lead capture.

• **Dead-Letter Queue Management:** Reprocesses failed or malformed payloads from integration endpoints.

Data Governance & Compliance

● Data Sovereignty

• **Deletion Approval Console:** Reviews soft-deleted lead requests in `DeleteRequestsPage.jsx` before authorizing permanent purge.

• **Bulk Data Export Controls:** Executes authorized enterprise data backups while preventing unapproved customer list exfiltration.

• **Comprehensive Audit Logs:** Audits `audit_logs` table tracking user logins, IP addresses, sensitive record updates, and permission changes.

• **Database Maintenance:** Oversees database backups, schema migration scripts, and indexing performance.

SECURITY & COMPLIANCE

21. Enterprise Security, Privacy & Data Governance

Defense-in-depth architecture safeguarding corporate customer intelligence

Authentication & Authorization

● Access Control

• **Stateless JWT Tokens:** Secure JSON Web Token authentication with expiration timestamps and signature verification.

• **Bcrypt Password Hashing:** Zero plain-text password storage; all credentials hashed with salt rounds via passlib/bcrypt.

• **Backend Route Guards:** FastAPI dependency injection (`require_permission`, `require_roles`) guarding every endpoint.

• **Database Query Scoping:** SQL queries automatically filter by employee ID or manager hierarchy at the database driver level.

API & Webhook Security

● Network & API Defenses

• **Meta HMAC-SHA256 Verification:** Validates Meta webhook signatures using SHA256 cryptographic keys to prevent payload spoofing.

• **Pre-Shared Header Secrets:** Website lead API protected by secret keys (`X-Website-Token`) and CORS origin policies.

• **Ghost Call Rejection:** Doocti webhook rejects internal station self-dialing loops and malformed caller data.

• **Nginx SSL & Rate Limiting:** All production traffic encrypted via TLS 1.3 with Cloudflare reverse proxy DDoS mitigation.

Data Sovereignty & Auditability

● Compliance Governance

• **Soft Delete Architecture:** Leads are never hard-deleted directly by reps; deletion requests require explicit Admin review.

• **Comprehensive Audit Logging:** Every edit, reassignment, status transition, and call record is written to immutable activity logs.

• **Controlled Data Exfiltration:** Strict permission checks on bulk Excel exports prevent unauthorized customer theft.

• **Privacy & Compliance:** Configurable for customer data deletion requests in alignment with digital data protection mandates.

OPERATIONAL REALITIES

22. Current Challenges & Implemented Improvements

Transparent engineering status: resolved bottlenecks, ongoing enhancements, and planned fixes

Resolved Enhancements

● **Completed & Live**

• **Ghost Call Blocker (Completed):** Engineered validation logic in `/api/doocti/call-log` to reject self-dialing PBX loops where `phone_number == station`.

• **Duplicate Lead Prevention (Completed):** Implemented phone sanitization and database uniqueness checks before lead creation.

• **Re-Inquiry Architecture (Completed):** Returning prospects are tagged as re-inquiries and attached to existing profiles without creating duplicates.

• **SOP Documentation (Completed):** Comprehensive documentation for Doocti, Proforma APIs, and Website Lead integrations published.

Active In-Progress Optimizations

● **In Progress (Current Sprint)**

• **Call Recording Sync (Active):** Enhancing webhook listener to poll or receive secondary Doocti recording-ready events when cloud transcoding is delayed.

• **WebSocket Auto-Reconnect (Active):** Strengthening frontend telephony WebSocket client to auto-reconnect gracefully after transient network drops.

• **Filter Performance (Active):** Adding compound database indexes on `(assigned_to, status, created_at)` to accelerate multi-parameter dashboard queries.

• **Meta Token Auto-Alert (Active):** Building automated notification when Facebook Page Access Tokens approach their 60-day expiration window.

Planned Technical Upgrades

● **Planned Infrastructure**

• **Async Webhook Message Queue:** Migrating webhook ingestion to Redis/Celery queue to decouple HTTP reception from database commits during heavy traffic spikes.

• **Automated Callback Dispatcher:** Auto-dialing missed inbound calls when the designated sales rep becomes available.

• **Advanced Reconnection Circuit:** Resilient retry circuit breakers for all external third-party API dependencies.

• **Real-time Push to Mobile:** FCM push notifications alerting field executives to new high-priority inbound inquiries.

STRATEGIC HORIZON

23 & 24. Future Roadmap & Executive Summary

Unifying marketing, telephony, sales execution, and intelligence into one platform

Strategic Product Roadmap (Next 2-4 Quarters)

- **Future Capabilities**

- **Native Android CRM App:** Full Flutter mobile client enabling field reps to manage leads, check follow-ups, and trigger SIM calls on the go.
- **Mobile SIM Telemetry Sync:** Privacy-compliant Android call duration and timestamp sync without violating Google Play background recording rules.
- **AI-Powered Predictive Lead Scoring:** Machine learning algorithms ranking inbound leads based on conversion probability, engagement, and profile data.
- **Automated WhatsApp Care Drips:** Configurable automated messaging journeys welcoming leads, sharing brochures, and collecting feedback.
- **Predictive & Power Dialing:** Automated outbound dialing campaigns connecting agents only when a live customer answers.
- **Integrated Revenue Forecasting:** Predictive revenue projections based on historical stage velocity and team quota achievement.

Unified Executive Value Summary

- **Platform Value Proposition**

- **Zero Lead Leakage:** Every lead from Meta Ads, Website forms, or referrals is captured instantly with 100% attribution.
- **Accelerated Speed-to-Lead:** Embedded 1-click Doocti telephony enables reps to connect with prospects within minutes of interest.
- **Complete Operational Accountability:** Objective telephony logs, call recordings, follow-up queues, and activity audit trails across all teams.
- **Structured Sales Governance:** Multi-tier RBAC protects customer data sovereignty while enabling managers to lead effectively.
- **Closed-Loop Revenue Attribution:** Directly connects marketing dollar spend to answered calls, submitted quotations, and closed-won revenue.
- **Enterprise Scalability:** Modern React + FastAPI architecture designed to scale seamlessly across hundreds of concurrent reps.